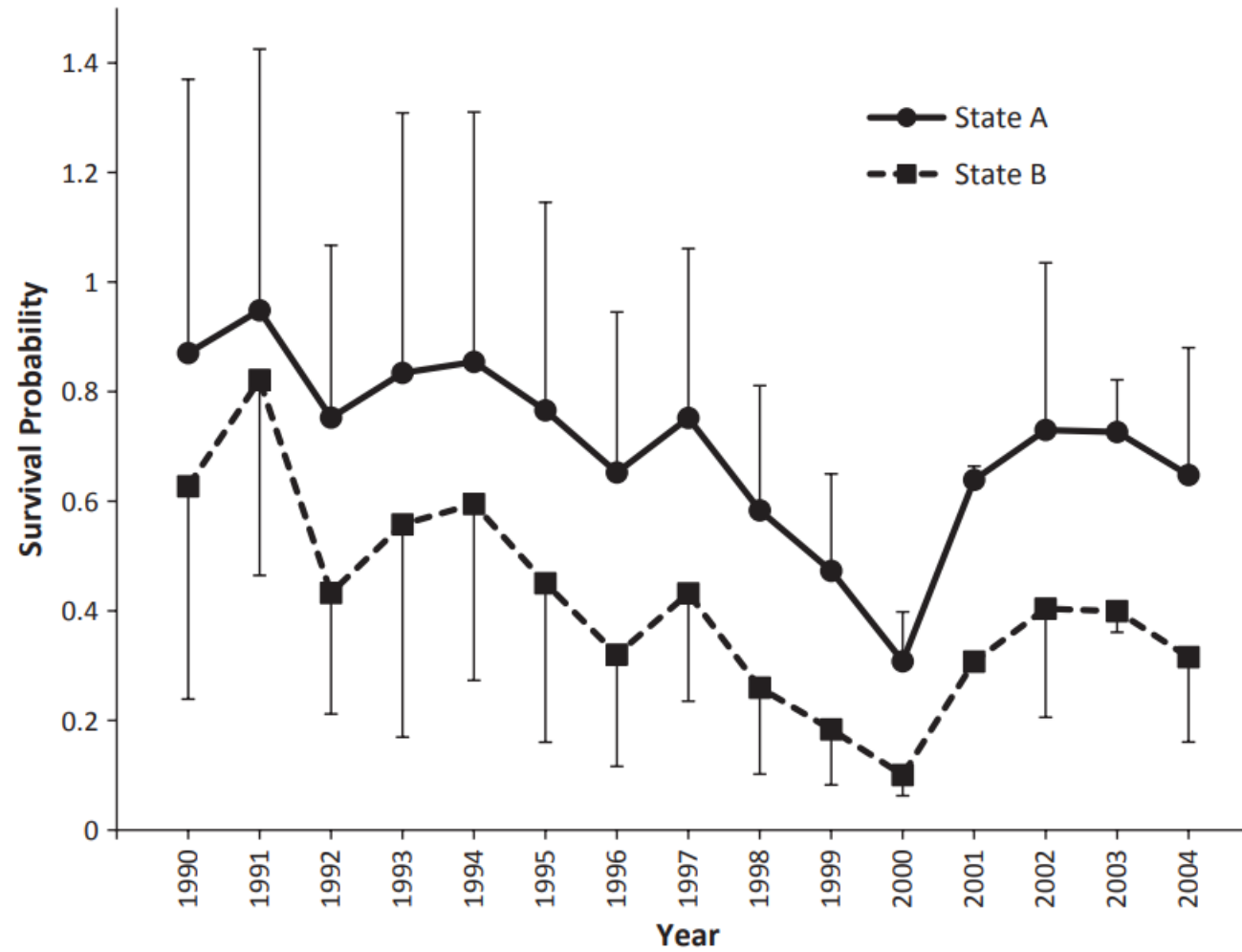


Link functions: constraining parameter space to the possible



But first, let's talk about 'assignments'

Project timeline

Date	Assignment
24 Sep	Topic summary
1 & 3 Dec	Oral presentations
10 Dec	Paper due

Topic Summary

- 1-2 pages
- Describe your project (very briefly, i.e., study species, research questions, covariates) and outline why (or why not) you wish to use a structural equation model to analyze your data
- It may be (very) helpful to include a 'causal diagram'

Causal diagram(s)

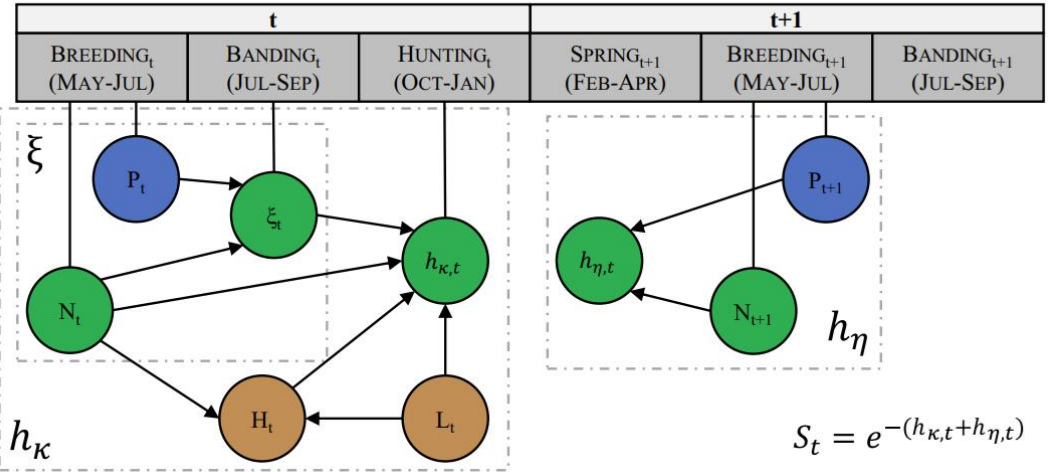


FIGURE 2 Directed acyclic graph demonstrating the hypothesized relationships among mallard breeding pair abundance (N), the number of ponds (P), harvest limits (L), the abundance of duck hunters (H), fecundity (ξ), harvest mortality hazard rate (h_{κ}), natural mortality hazard rate (h_{η}) and survival (S) for mallards marked and released in the Prairie Pothole Region of the United States and Canada, 1974–2016. Arrows represent covariate effects, grey dashed lines enclose separate generalized linear models and vertical solid lines denote the time period or interval when parameters were estimated, where survival (S) and natural mortality in year t are estimated from banding in year t to banding in year $t+1$. We estimated age-specific band recovery probabilities (f) as a function of age-specific harvest probability (κ), reporting rate (r) and crippling rate (c), $f = \kappa(1 - c)r_t$. We note that we hypothesized the same relationships among demographic components for both juvenile and adult females.

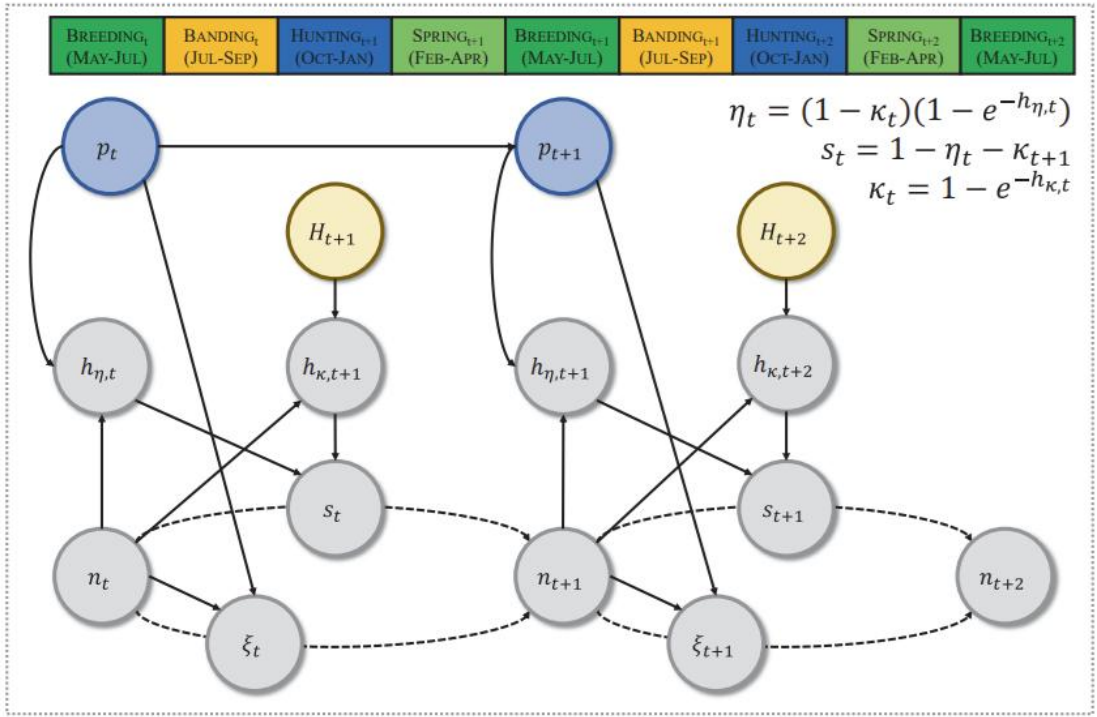


FIGURE 2 A directed acyclic graph demonstrating the relationships among abundance (n), ponds (p ; blue), fecundity (ξ), hunting mortality hazard rate (h_{κ}), natural mortality hazard rate (h_{η}), survival (s) and the number of duck hunters (H ; brown) for blue-winged teal breeding in the North American Prairie Pothole Region across the annual cycle (1973–2016). Solid arrows represent estimated directional relationships, and dashed arrows represent processes leading to changes in population abundance.

Causal diagram(s)

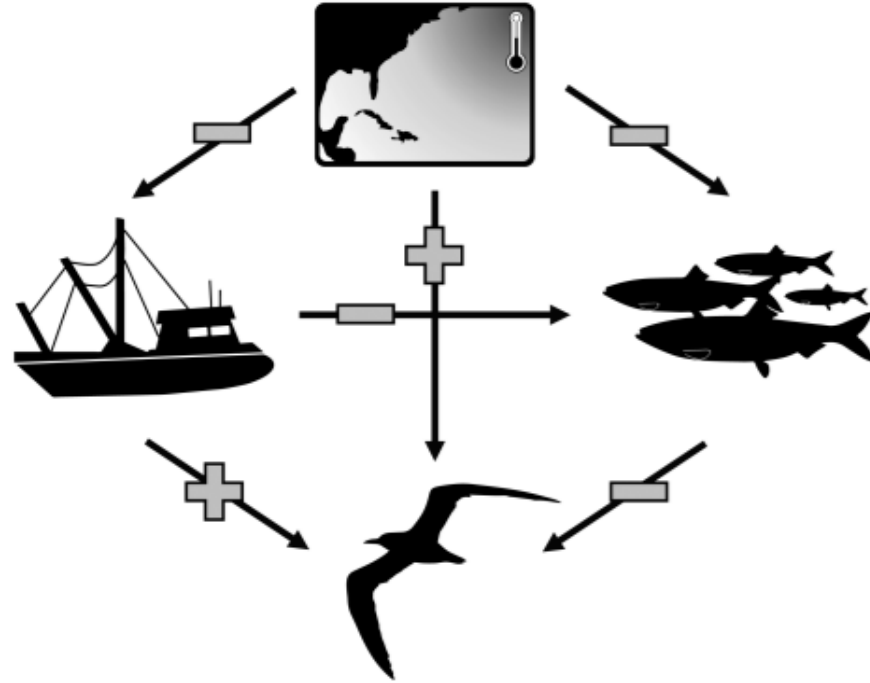
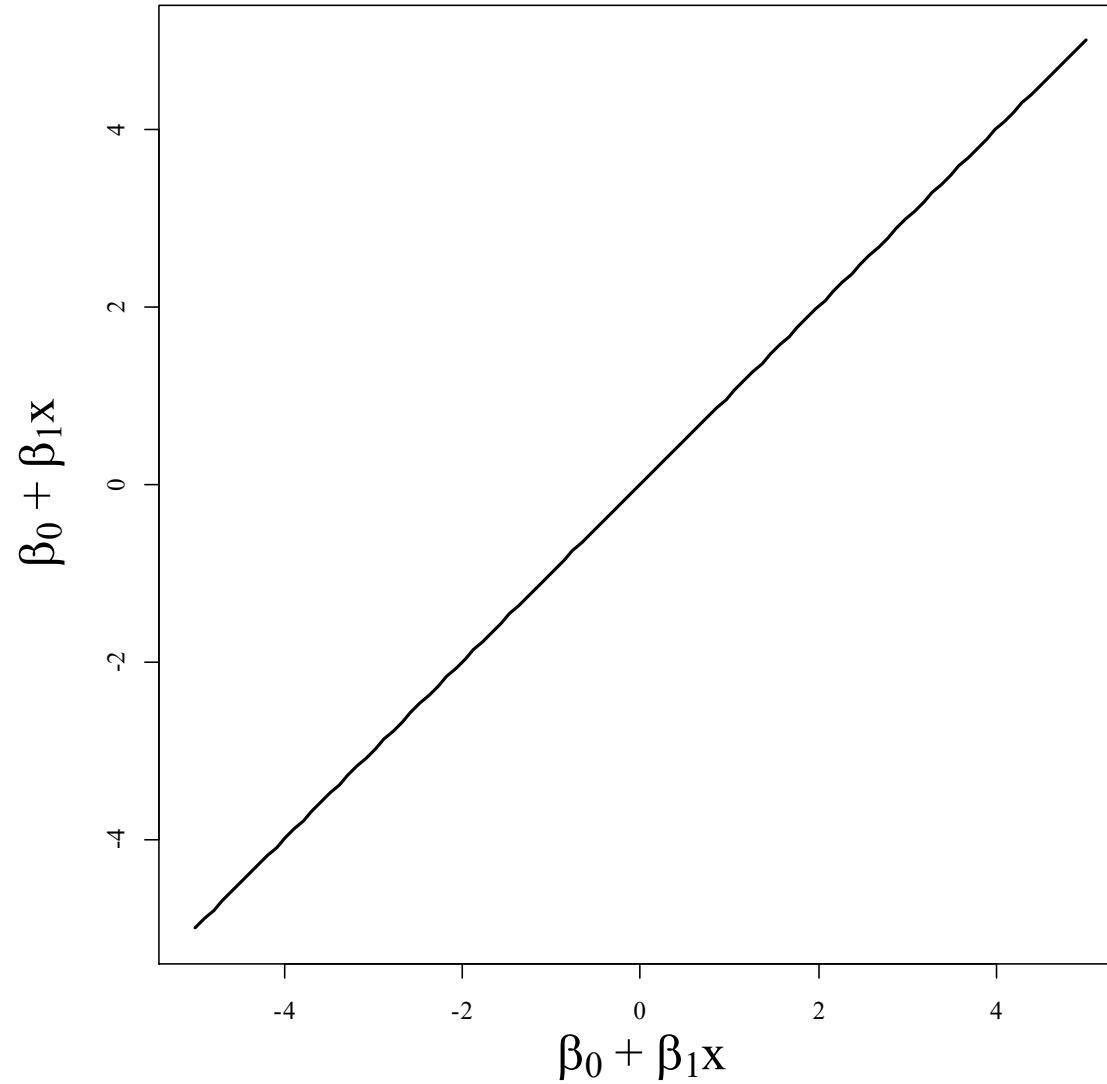


FIGURE 1 Simplified path diagram describing the hypothesized directionality (plus: Positive, bar: Negative association) regarding how environmental variables (i.e., sea-surface temperatures in the North Atlantic, fishery pressure, and fish production) influenced one another, as well as the indirect and direct pathways in which these sources of environmental variability influenced Royal tern mortality.

Thus far we've discussed linear models

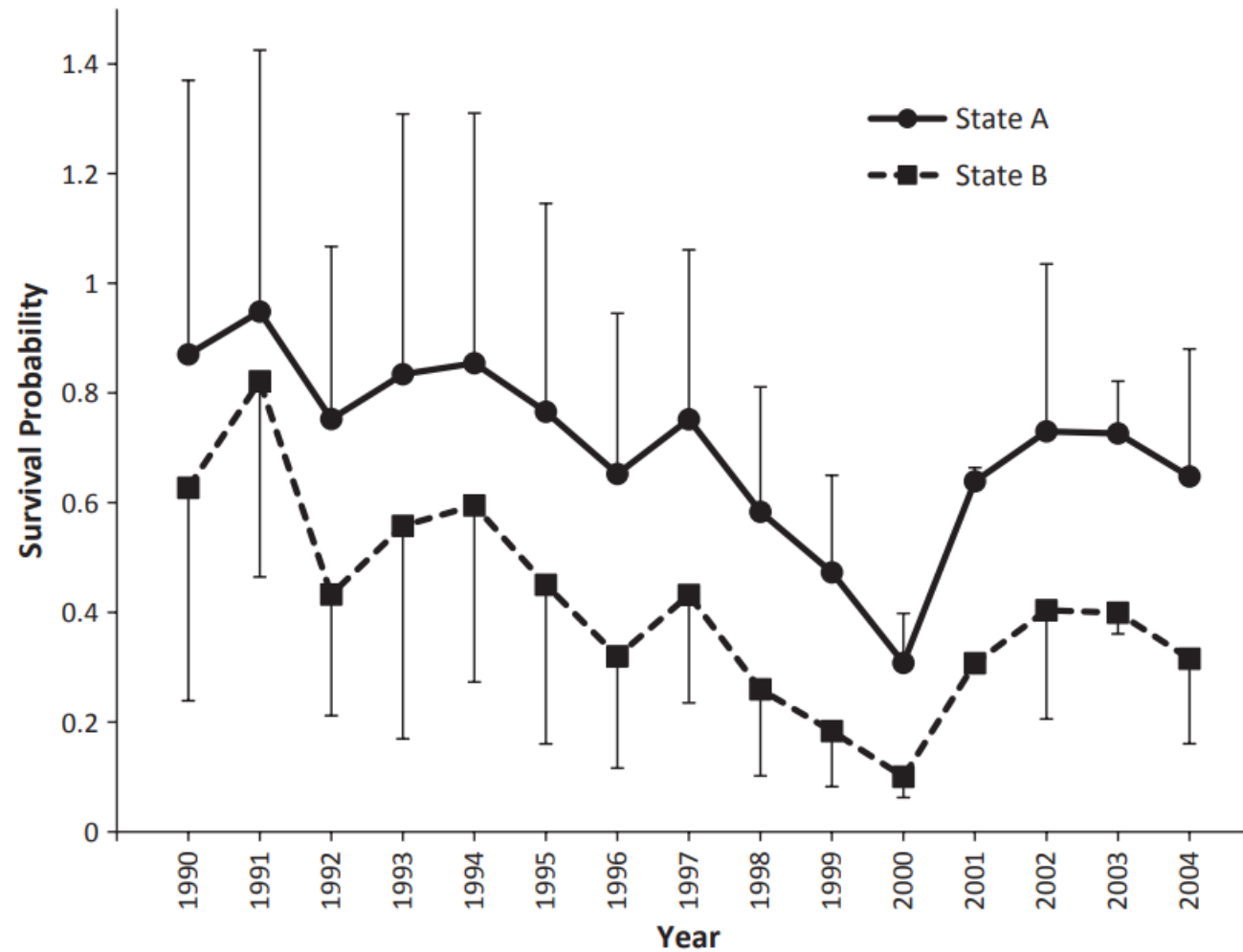
Linear models use an 'identity link'

The scale of our response variable $[-\infty, \infty]$



The scale of our model $[-\infty, \infty]$

Transformations are often necessary [take a close look]



Lindberg [43, 4755], Sedinger [53; 7668], and Lebreton [76; 22357] **(2013)** *Ecology and Evolution*

Transformations are often necessary

How do we prevent ourselves from saying crazy [stuff]?

Let's imagine some ecological data types

Let's imagine some ecological data types

1. Counts of individuals [discrete, never less than 0?!]
2. Lengths, weights, or other morphometric measurements [continuous, never less than 0?]
3. Number of eggs or offspring [discrete, never less than 0]

Let's imagine some ecological data types

1. Proportion or percent cover of something or other
2. Survival [yes/no; 1/0]
3. Pregnancy [yes/no; 1/0]
4. Hatching [yes/no; 1/0]
5. Morph [categorical?; 1/0 or 1, 2, ... , n]

All of these data types share some common attributes

1. You can't have < 0 's
2. Many can only be 0's or 1's

We often need to constrain parameters to be > 0 or between 0 and 1.

So how do we handle that in our models?

Link functions!

How the heck do we do that?

Link functions: constraining parameter space to the possible



Or, what is a natural log?

What is e ? [an irrational and transcendental number]

e is the basis of many of the link functions used in Ecology

Link functions allow us to constrain parameter estimates so we don't predict things like 150% survival, 200% shrub cover, or that an alligator that is -2 ft long will weigh -100 pounds

M. S. Lindberg *et al.*

Heterogeneity and Harvest Dynamics

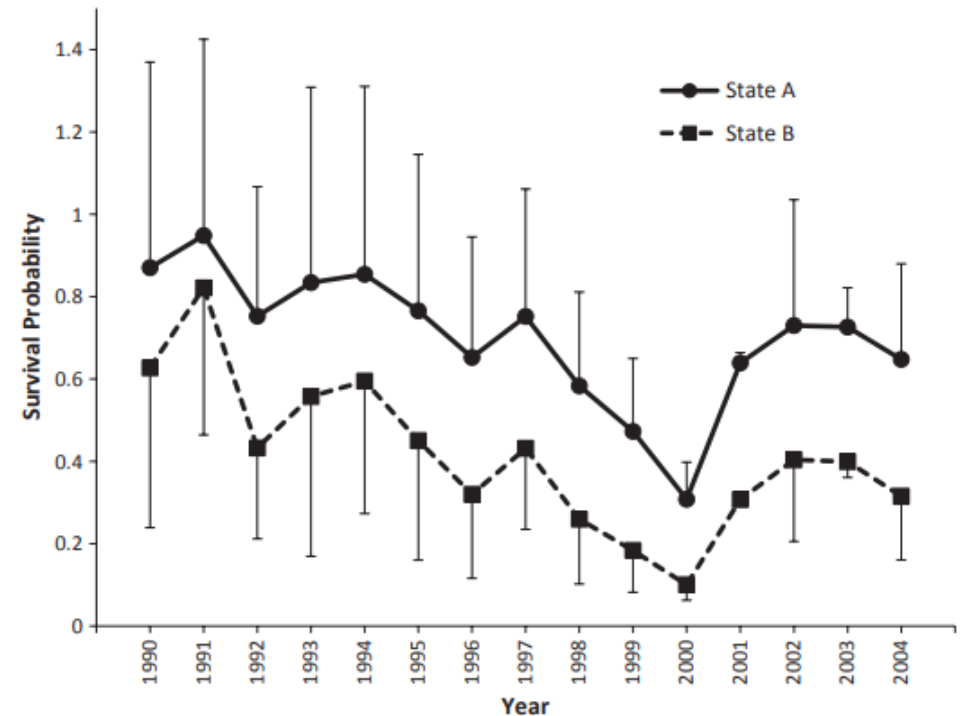


Figure 2. First-year survival probability of brant for states A and B. Errors bars are ± 1 standard error in single direction for clarity.

What is e? [an irrational and transcendental number]

Imagine you have a bank account with \$1 in it that earns 100% interest [put more money in the account!]

If you calculate interest annually, at the end of the year, you'll have \$2

$$\$1 \times 2^1 = \$2$$

What is e? [an irrational and transcendental number]

Imagine you have a bank account with \$1 in it that earns 100% interest [put more money in the account!]

If you calculate interest biannually, at the end of the year, you'll have \$2.25

$$\$1 \times 1.5^2 = \$2.25$$

What is e? [an irrational and transcendental number]

Imagine you have a bank account with \$1 in it that earns 100% interest [put more money in the account!]

If you calculate interest quarterly, at the end of the year, you'll have \$2.44

$$\$1 \times 1.25^4 = \$2.44$$

What is e ? [an irrational and transcendental number]

Compounding annually yields 2

Compounding biannually yields 2.25

Compounding quarterly yields 2.4414...

Compounding monthly yields 2.613035....

Compounding weekly yields 2.692597...

Compounding daily yields 2.714567...

Notice a pattern?

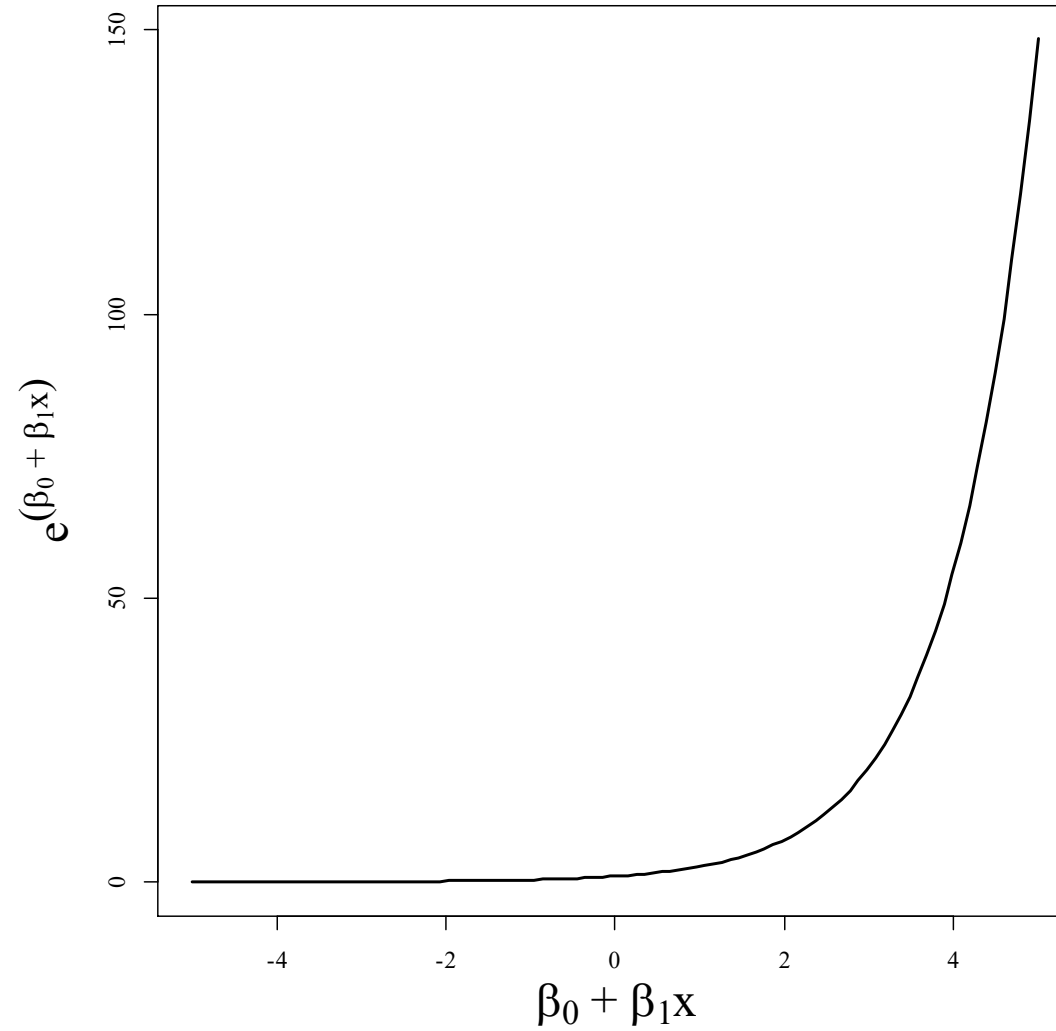
$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$

Compounding **continuously** yields e

2.71828182845904523536028747135266249775724709369995957496696762772
407663035354759457138217852516642742746639193200305992181741359662
904357290033429526059563073813232862794349076323382988075319525101
901157383418793070215408914993488416750924476146066808226480016847
741185374234544243710753907774499206955170276183860626133138458300
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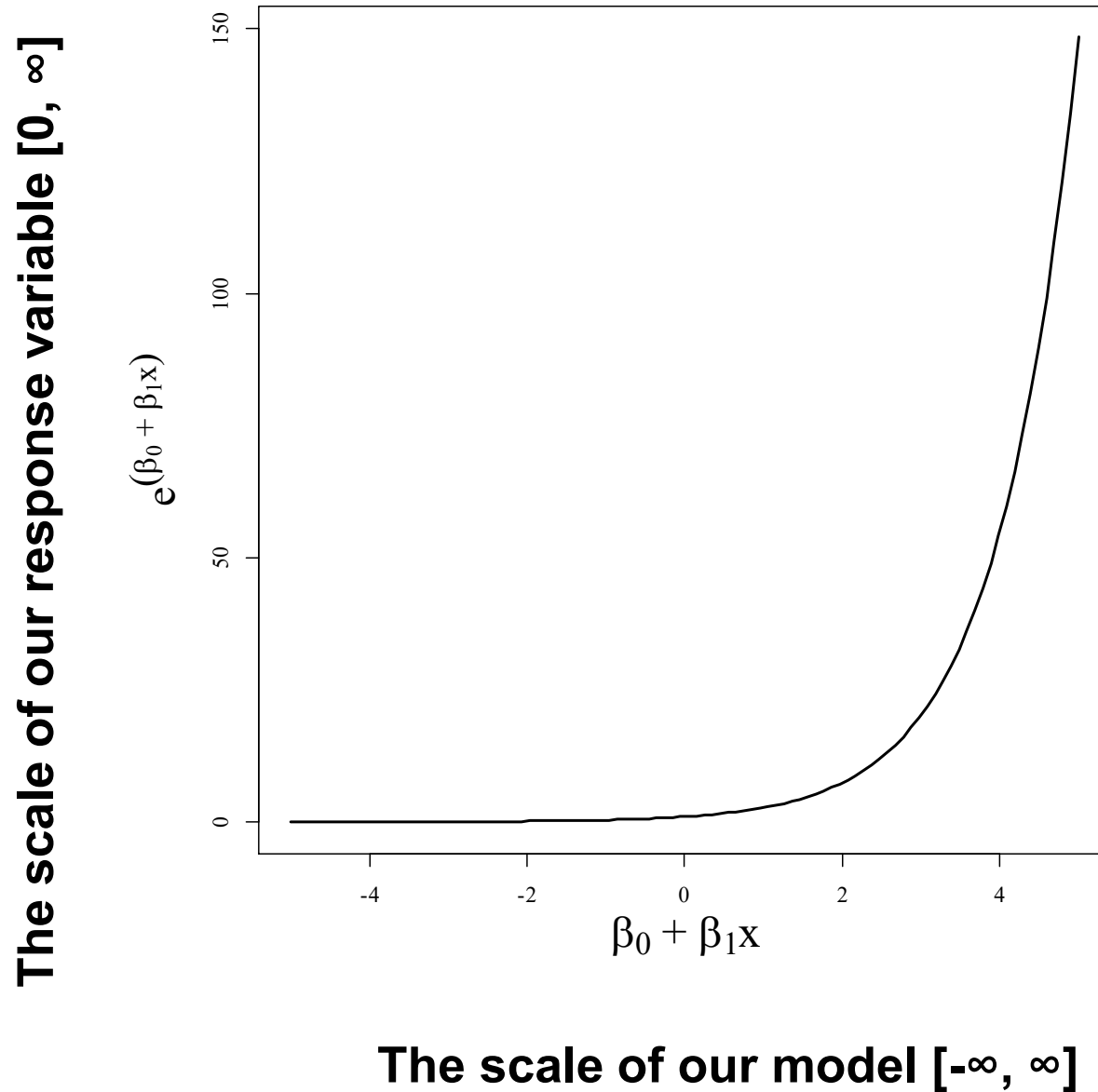
We can do beautiful things with e (constrain to > 0)

The scale of our response variable $[0, \infty]$



The scale of our model $[-\infty, \infty]$

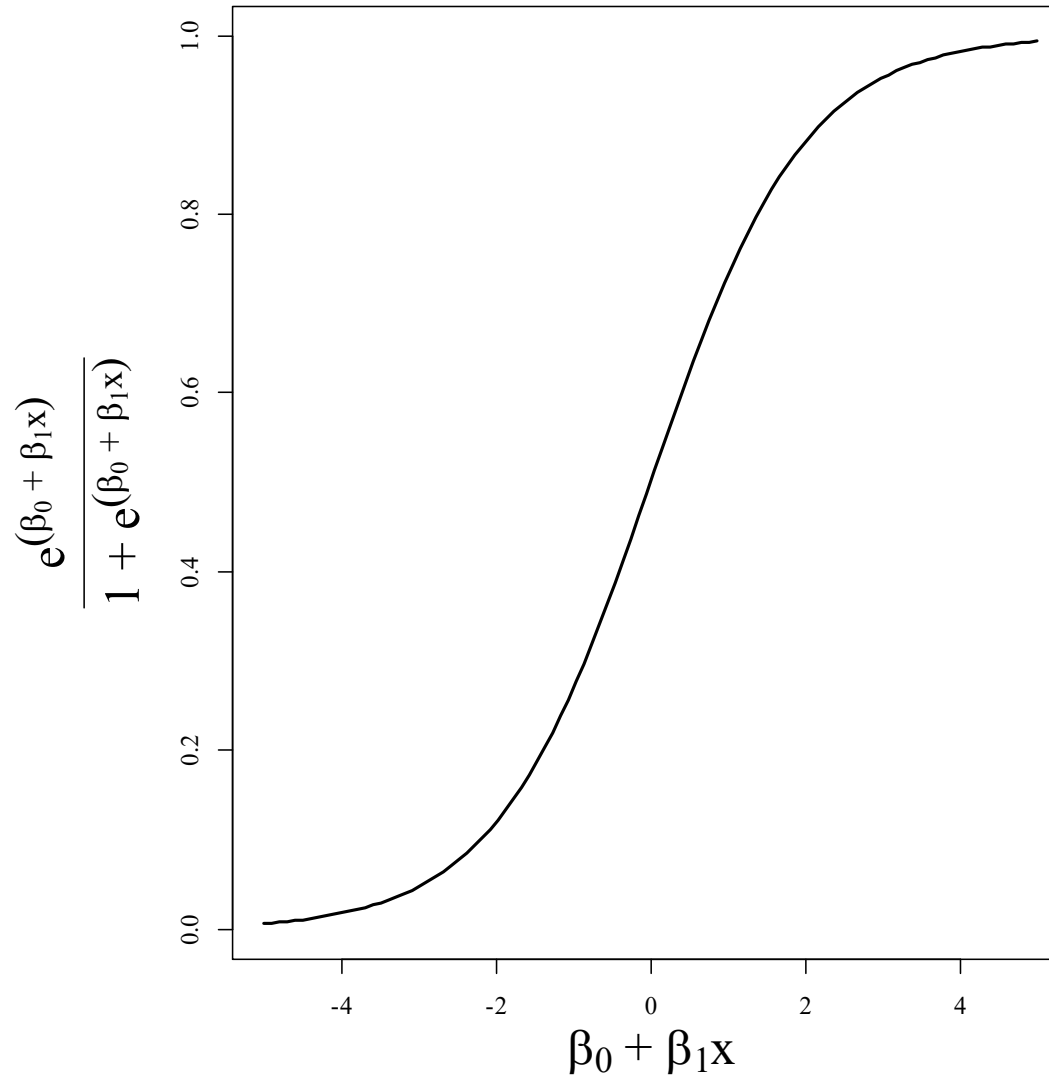
We can do beautiful things with e (constrain to > 0 & < 1)



$$y = e^{\beta_0 + \beta_1 \times x}$$
$$\beta_0 + \beta_1 \times x = \ln(y)$$

We can do beautiful things with e (constrain to > 0 & < 1)

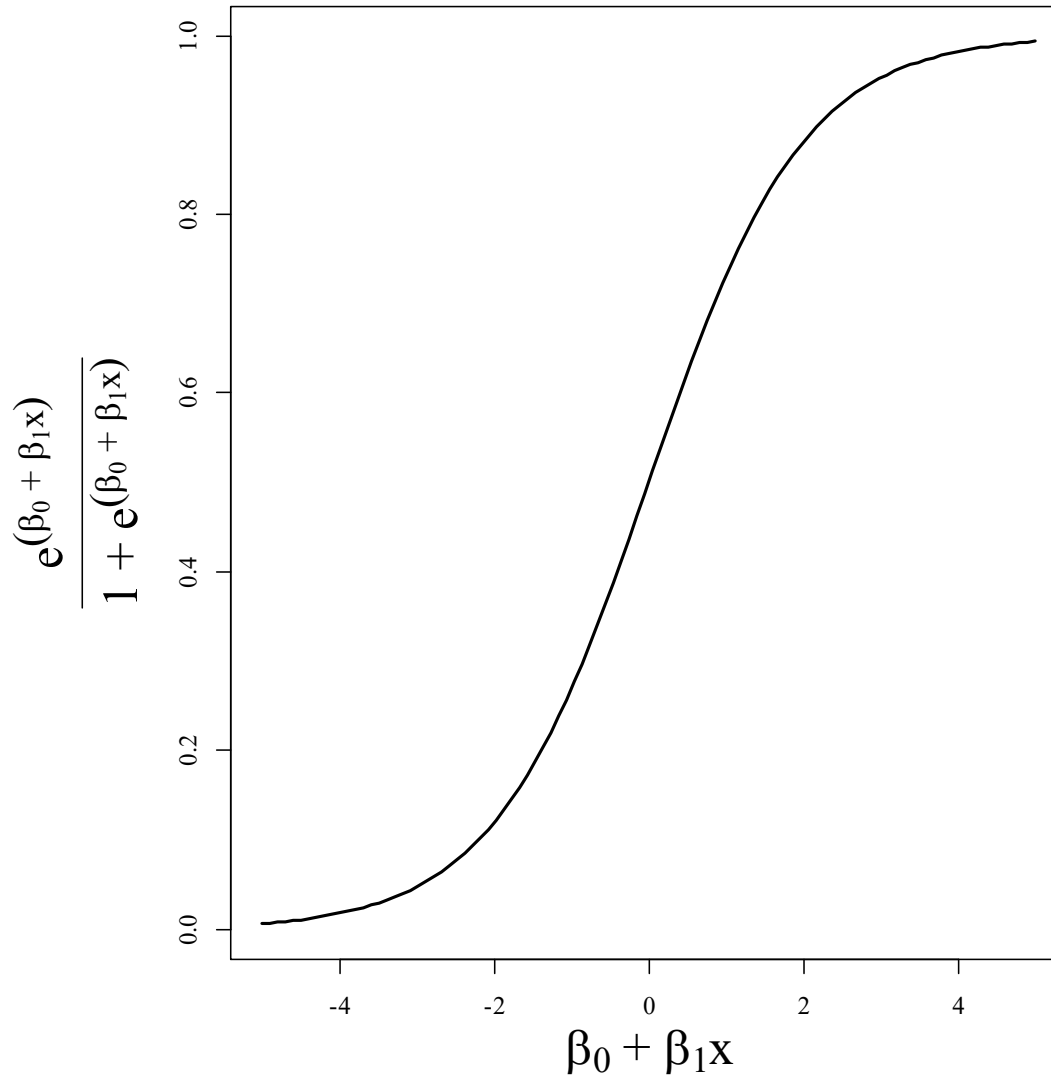
The scale of our response variable $[0, 1]$



The scale of our model $[-\infty, \infty]$

We can do beautiful things with e (constrain to > 0 & < 1)

The scale of our response variable [0, 1]

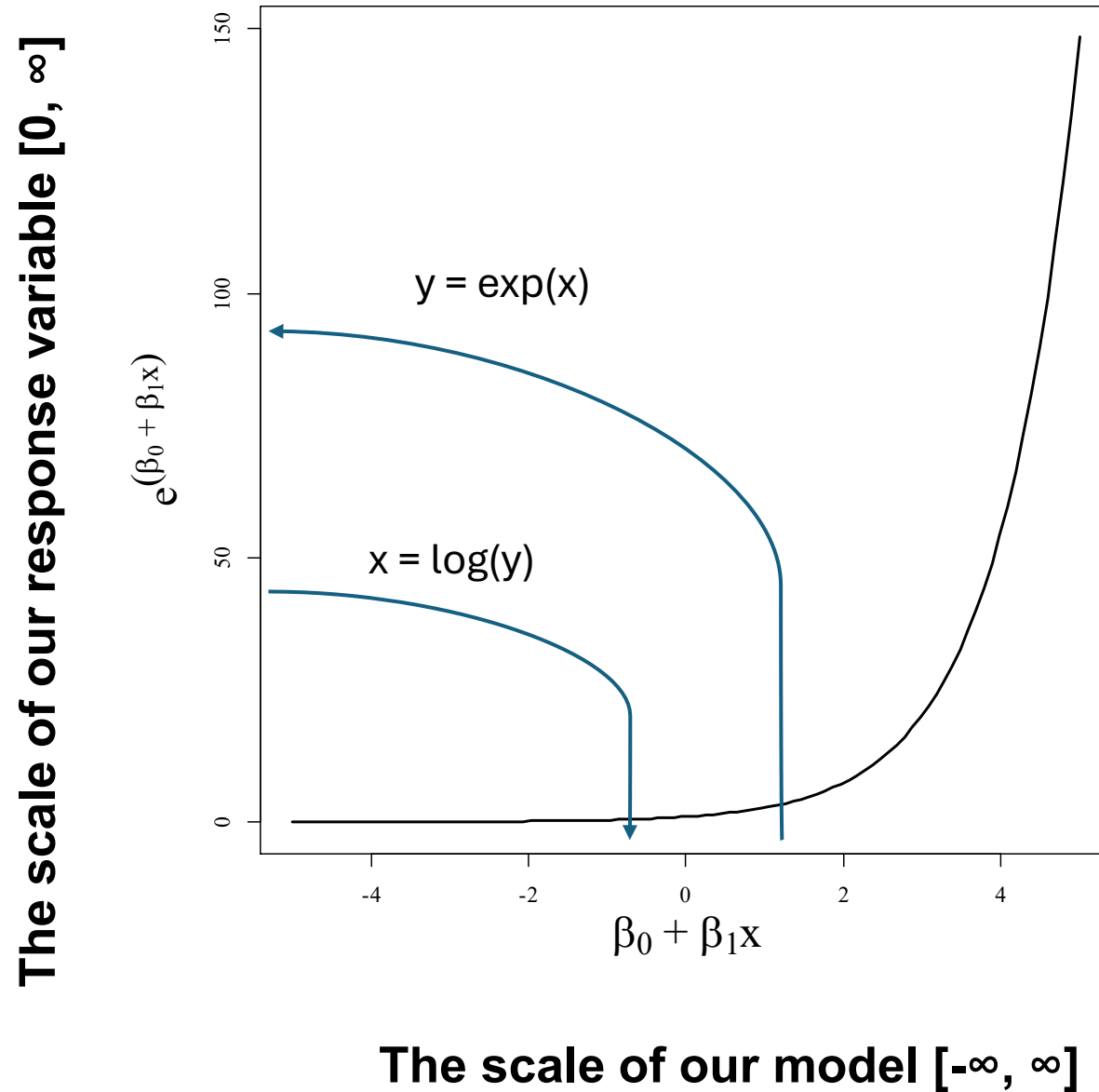


The scale of our model $[-\infty, \infty]$

$$y = \frac{e^{\beta_0 + \beta_1 \times x}}{1 + e^{\beta_0 + \beta_1 \times x}}$$

$$\beta_0 + \beta_1 \times x = \ln\left(\frac{y}{1 - y}\right)$$

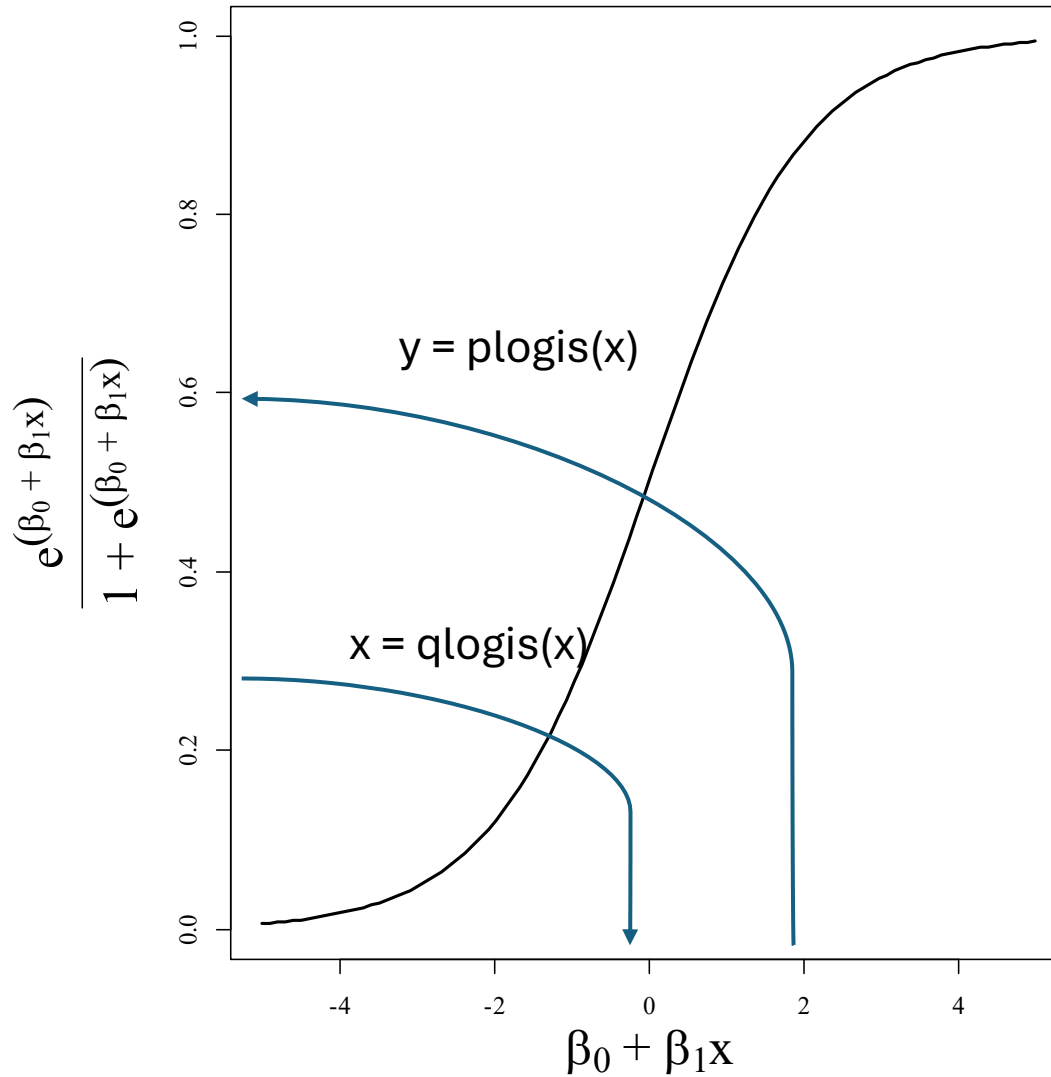
Implementing in base R (log link)



$$y = e^{\beta_0 + \beta_1 \times x}$$
$$\beta_0 + \beta_1 \times x = \ln(y)$$

Implementing in base R (logit link)

The scale of our response variable [0, 1]



The scale of our model $[-\infty, \infty]$

$$y = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

$$\beta_0 + \beta_1 x = \ln\left(\frac{y}{1 - y}\right)$$

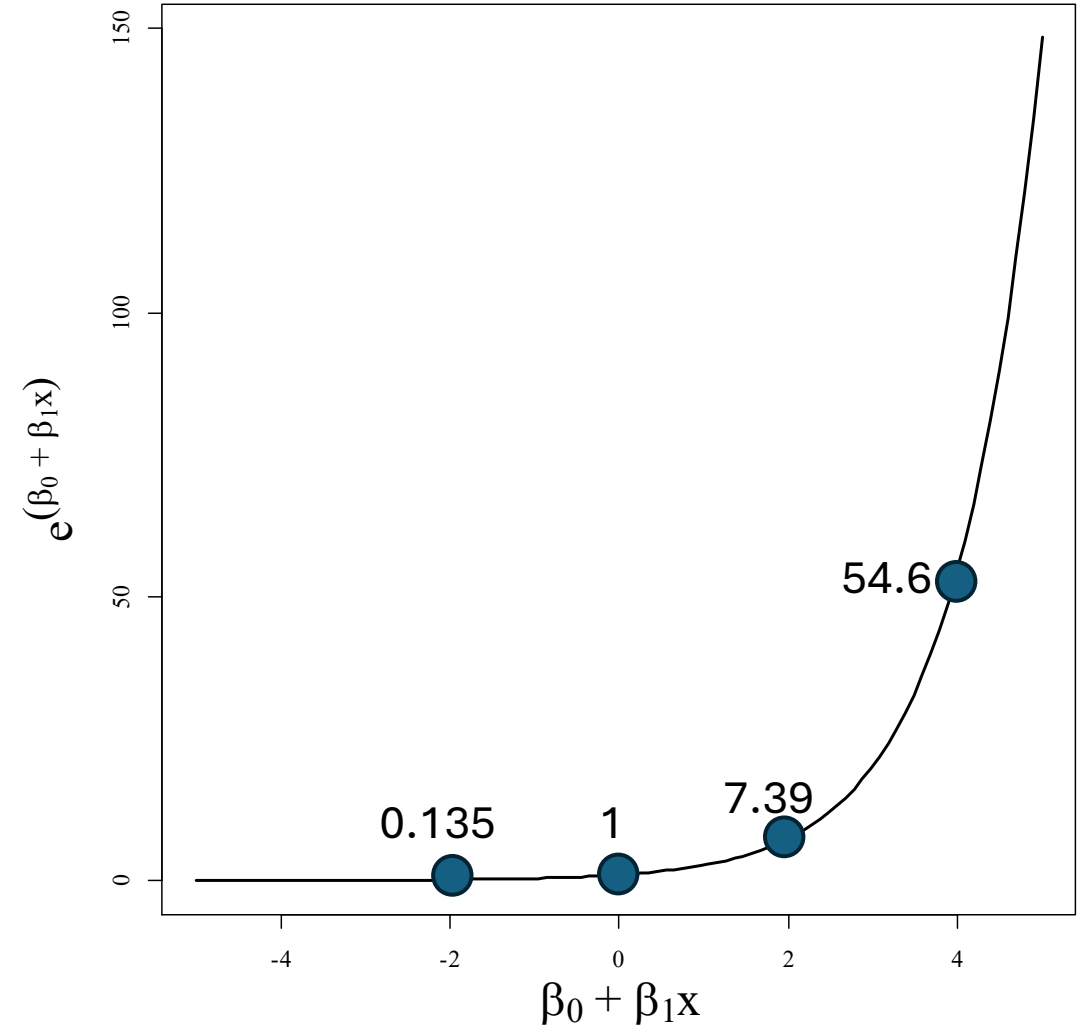
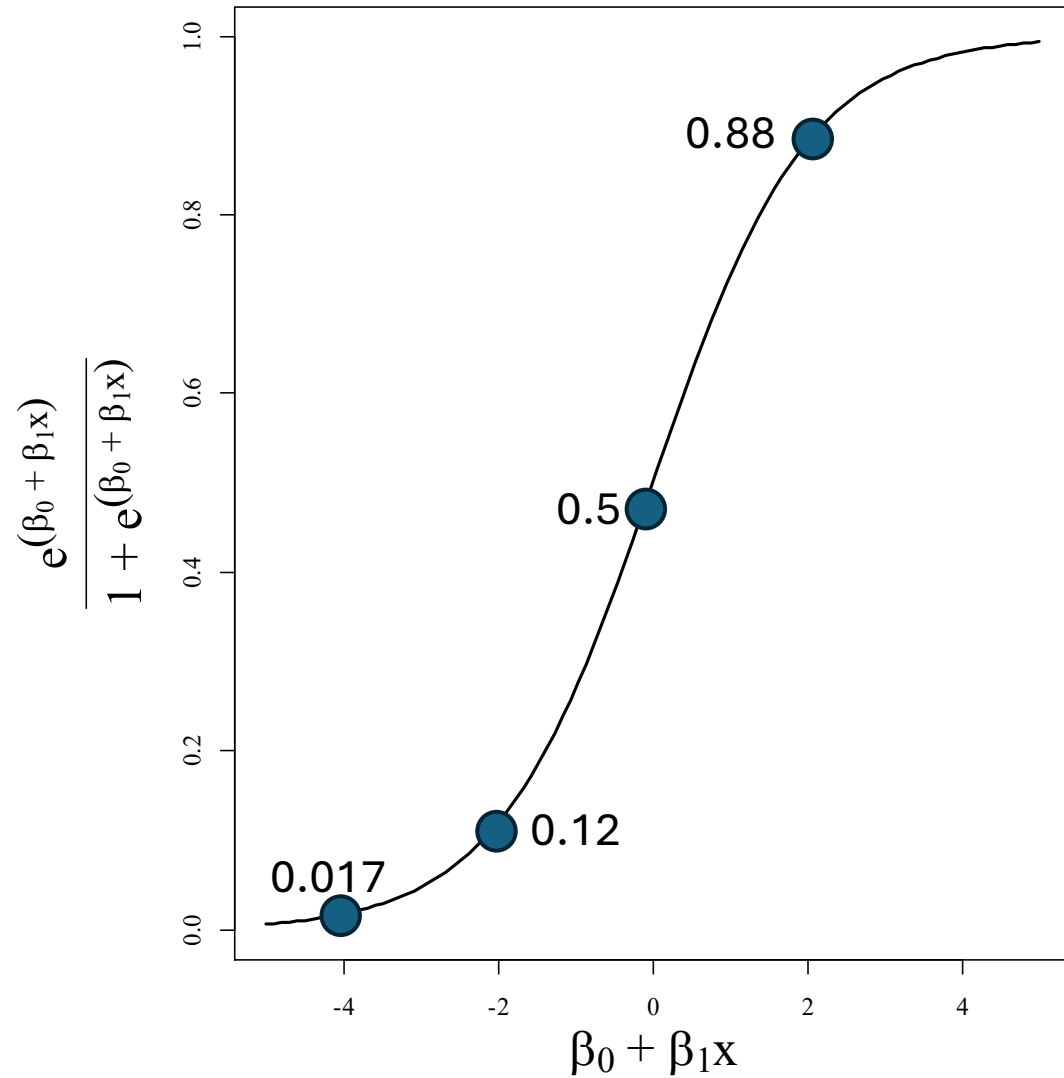
Today, we'll reanalyze two datasets...

1. We'll model the alligator mass ~ length data using a log-normal distribution (i.e., constraining mass to be positive!)
2. We'll model some duck band-recovery data using a logit-link (i.e., constraining the probability of being shot, recovered, and reported to be > 0 and < 1)

Things to think about:

1. Beta estimates are no longer 'cut and dry' rise/run. They are relative to your position on the link function and the shape of the link function.

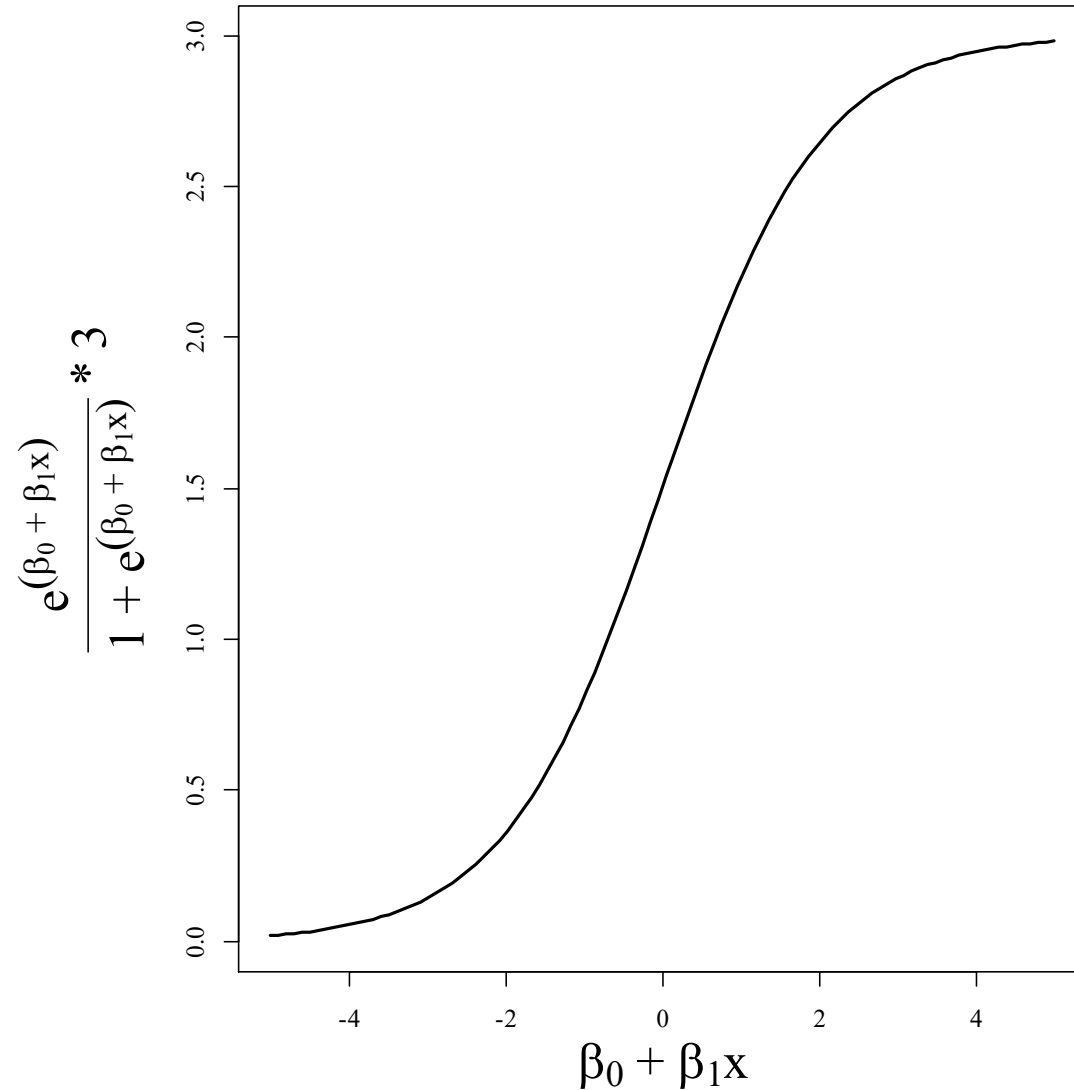
Beta estimates are no longer cut and dry!



Things to think about:

1. Beta estimates are no longer 'cut and dry' rise/run. They are relative to your position on the link function and the shape of the link function.
2. There are many, many different types of link functions that do very similar things. Sometimes using a slightly different function can help with parameter estimation (related to point #1).

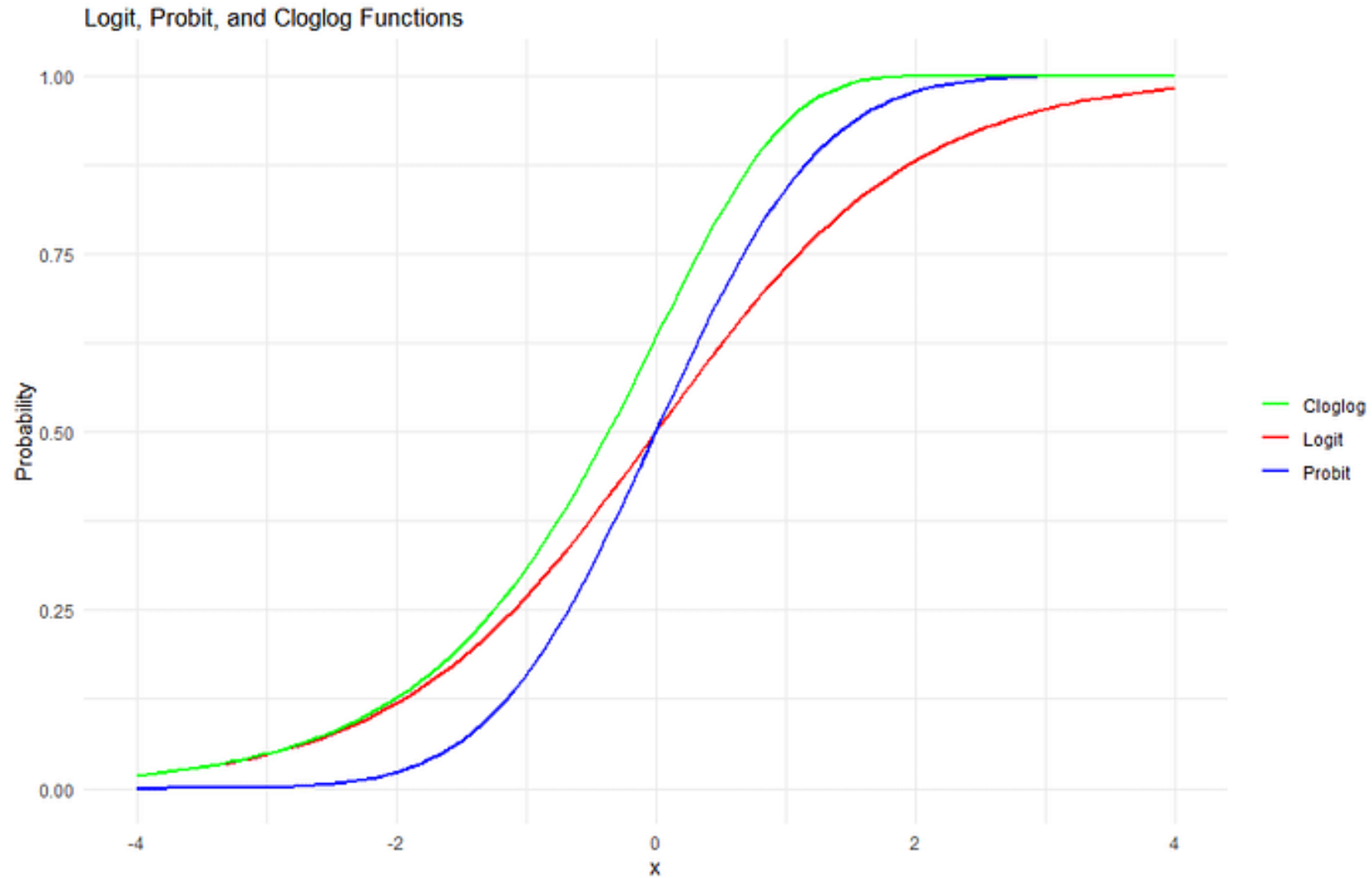
Different link functions (an example to constrain b/w 0 and 3)



$$y = 3 \times \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

$$\beta_0 + \beta_1 x = \ln \left(\frac{\frac{y}{3}}{1 - \frac{y}{3}} \right)$$

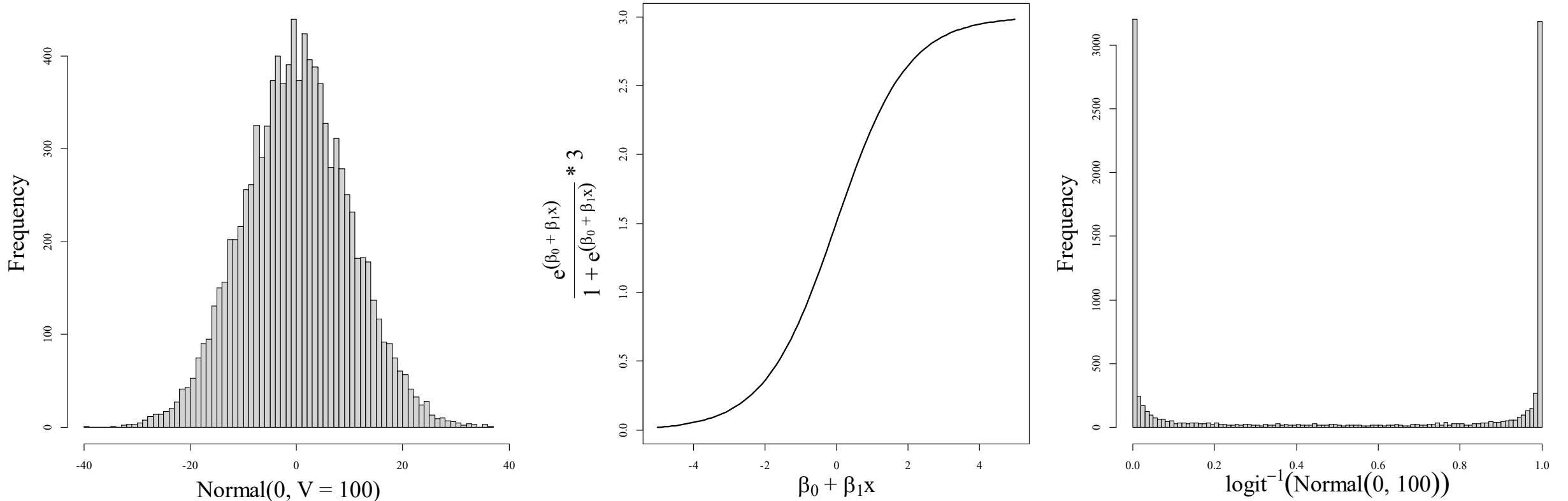
Different link functions (different shapes)



Things to think about:

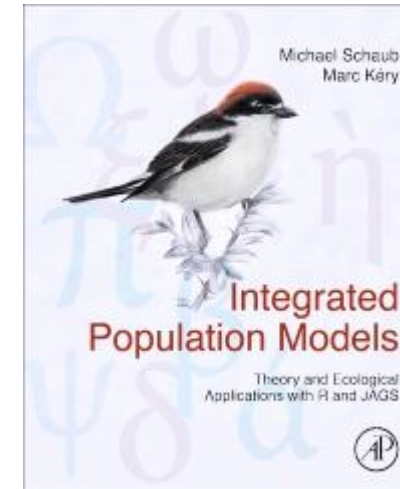
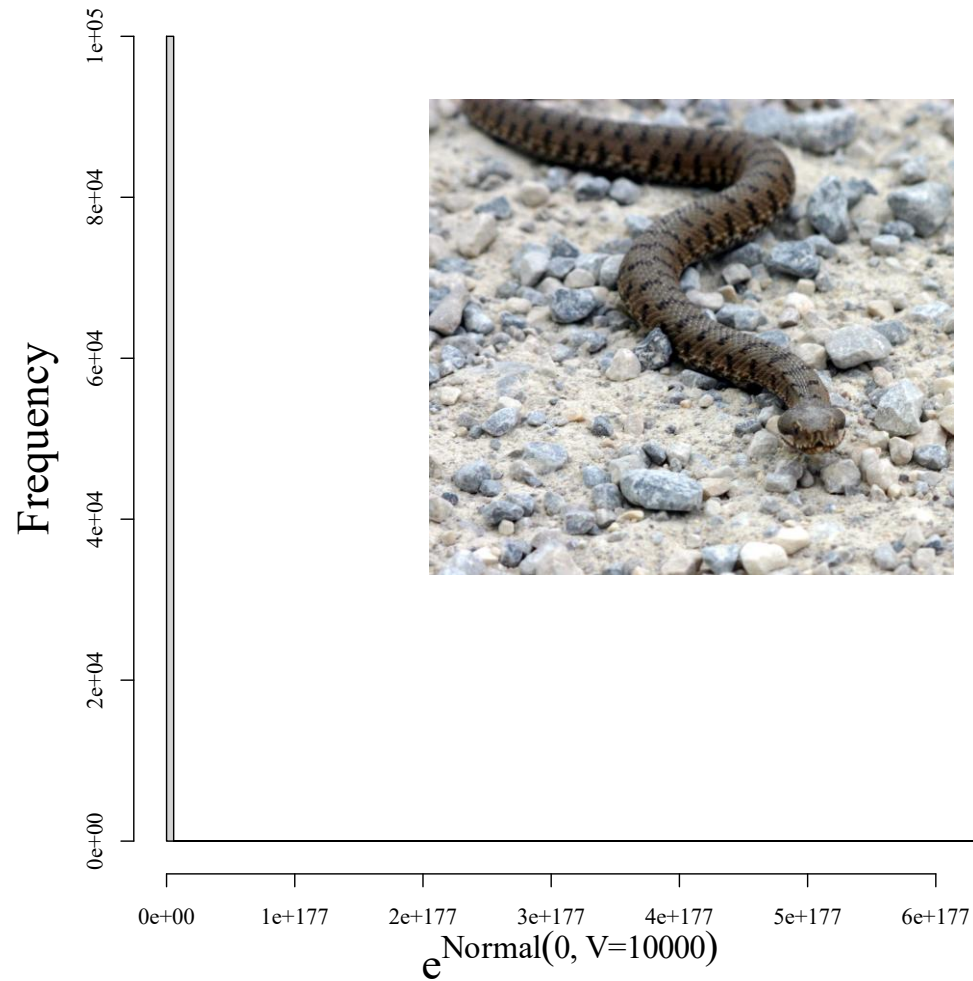
1. Beta estimates are no longer 'cut and dry' rise/run. They are relative to your position on the link function and the shape of the link function.
2. There are many, many different types of link functions that do very similar things. Sometimes using a slightly different function can help with parameter estimation (related to point #1).
3. Priors suddenly become **quite a bit more complex** to think about... (see Northrup and Gerber 2018 for a logit-link example... and Lecture 2 for some log-link examples, e.g., snakes larger than a solar system!)

Priors on link functions



Northrup & Gerber (2018) *PLOS ONE*

Priors on link functions

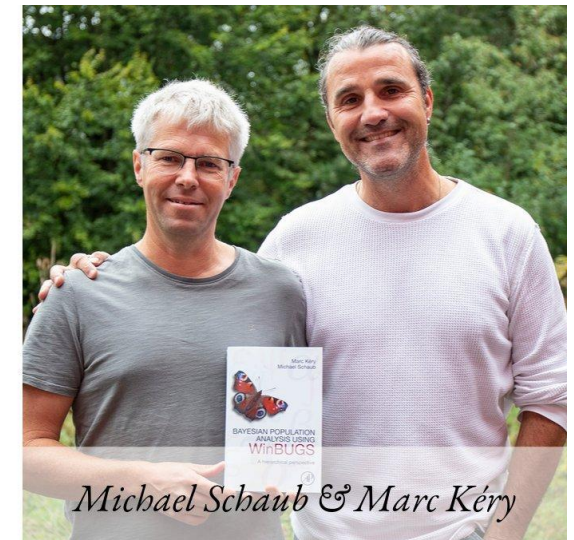


The Marsh Award for **INNOVATIVE ORNITHOLOGY**

Nominated

Michael and Marc were nominated for their groundbreaking work on Bayesian hierarchical models, changing the way we use statistics to analyse large, citizen-science data sets. The methods have not only helped BTO, but are being used worldwide on a variety of data sets and applications. Their books, as well as their workshops and teaching, only adds value to their work.

MARSH
Christian Trust



Michael Schaub & Marc Kéry

Earth's diameter is 12,756 km